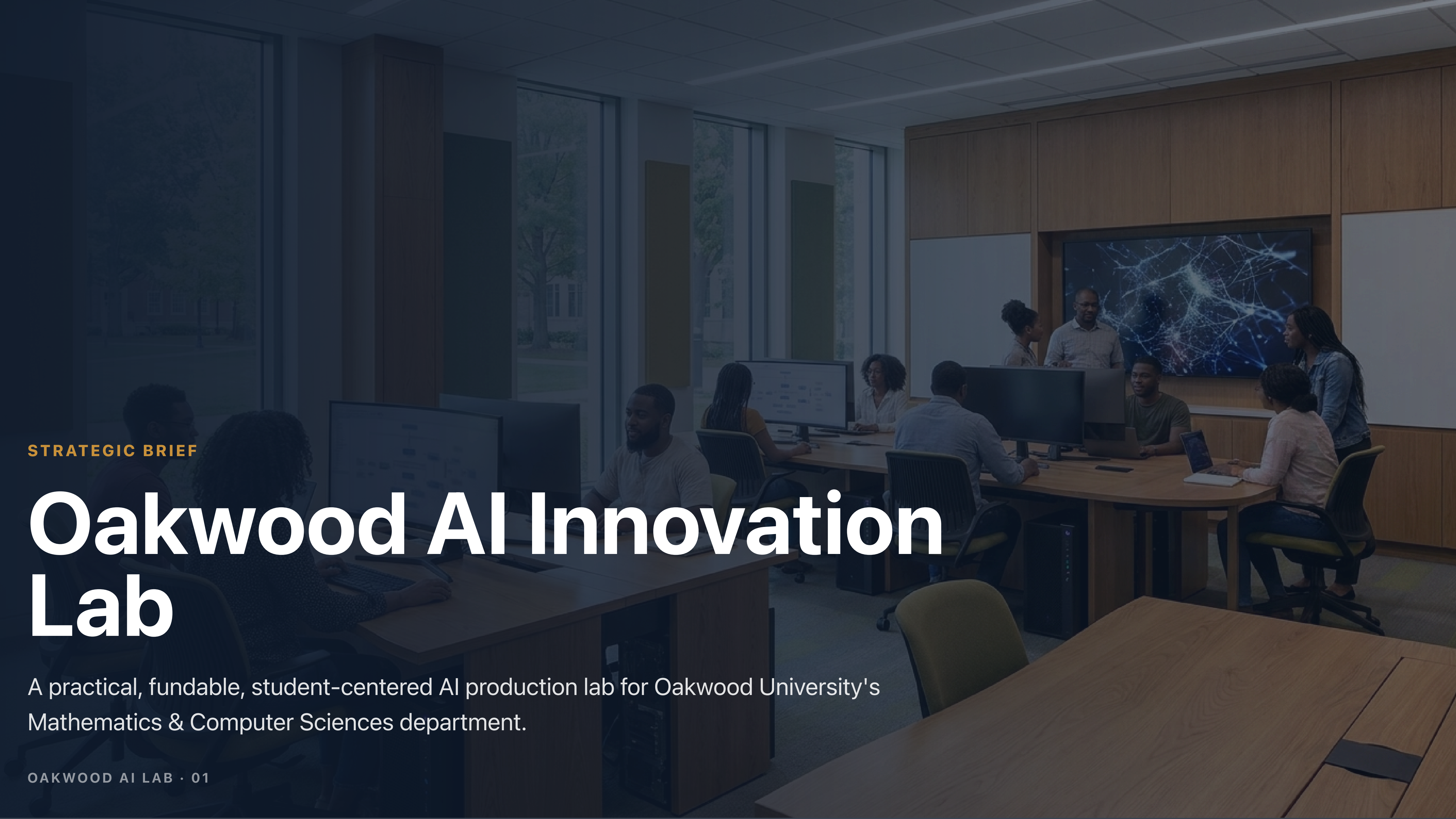


STRATEGIC BRIEF

Oakwood AI Innovation Lab

A practical, fundable, student-centered AI production lab for Oakwood University's Mathematics & Computer Sciences department.

EXECUTIVE THESIS

Do not build a computer lab. Build a production studio.

The opportunity is not simply new desktops. The opportunity is a lab where students use AI, build AI systems, run local models, practice governance, and leave with public portfolio work.

- 10-seat lab concept designed around student output, not hardware display.
- Four budget paths: $20K, $50K, $100K, and $200K.
- Hardware, curriculum, governance, and deployment are planned together.
- The recommended next move is to confirm the funding window and produce a procurement-ready memo.

WHY NOW

AI has compressed the distance between idea and implementation.

High school and college-age builders are already launching apps, media systems, and AI-assisted products at professional speed. Oakwood can turn that reality into a structured advantage for its students.

- Students should graduate with shipped projects, demos, and technical explanations.
- Huntsville's defense, aerospace, cyber, and engineering ecosystem gives the lab a natural employer pipeline.
- HBCU AI capacity is a national workforce priority; Oakwood can frame this as workforce readiness plus ethical leadership.
- Local model capability matters because privacy, security, latency, cost, and governance cannot be learned only through cloud chat tools.

OAKWOOD POSITION

HBCU + Adventist + STEM is a distinct AI voice.

Oakwood can speak to responsible AI from a position most institutions cannot: technical skill joined to human dignity, service, stewardship, and community trust.

- The lab can serve Computer Science, Applied Math, Information Technology, Business Analytics, Media, Health, Theology research, and campus operations.
- Responsible AI becomes a practiced workflow: data classification, approval gates, audit trails, evaluation, and human-in-the-loop design.
- A visible lab gives donors, administrators, faculty, and students a concrete artifact to rally around.

LEARNING MODEL

Four pillars: use AI, build AI, run local, govern AI.

Each pillar maps to classroom workflows, physical lab zones, and portfolio-ready student outcomes.

Use AI

Prompting, research, writing, analysis, design, coding, and presentation.

Build AI

Apps, agents, retrieval systems, dashboards, voice, vision, and automations.

Run Local

Ollama/vLLM, quantization, GPU constraints, model serving, and latency tradeoffs.

ROOM CONCEPT

A 10-seat lab with six functional zones.

The room should visibly communicate that students are building, testing, presenting, and governing AI systems.

- Student build stations
- Local model cluster
- Demo wall
- Studio corner

HARDWARE ARCHITECTURE

Use a mixed stack. One machine type cannot teach the whole field.

The lab should include everyday student seats, one or more local AI nodes, shared GPU capacity where budget allows, edge AI devices, and serious storage/networking.

Student Seats

Framework Desktop AI Max, Mac Studio, or GPU PCs for daily production.

BUDGET PATHS

Four complete configurations, not four shopping lists.

Each tier includes hardware, lab purpose, curriculum use, and the next deliverable package.

$20K

Starter Pod

Pilot lab: 4-6 capable seats, shared local AI node or edge kits, display, open-source stack.

$50K

Production Lab

Recommended minimum: 6-10 strong seats, DGX Spark-class node, demo wall, NAS/network, 8-week pilot.

$100K

RECOMMENDATION

The $50K tier is the best first move; $100K is the credibility jump.

If the available funding is uncertain, start with the $50K production lab because it creates visible student output quickly. If the department has real grant or donor capacity, the $100K research lab is the stronger institutional signal.

- $20K proves interest but will feel constrained fast.
- $50K can launch a credible student production lab this semester.
- $100K supports faculty research, employer engagement, and a stronger grant story.
- $200K should be framed as a donor/naming opportunity, not the first ask unless capital is already available.

CURRICULUM SPINE

Eight modules turn the room into outcomes.

The curriculum should be modular enough for a course, club, summer bridge, certificate, or faculty development track.

AI Fluency

Prompting, source discipline, academic integrity, multimodal workflows.

Build Week

GitHub, Vercel, databases, auth, and AI-assisted app development.

Local Models

Embeddings, RAG, quantization, local inference, and model selection.

GOVERNANCE

Responsible AI should be practiced, not merely stated.

A strong lab gives students power and teaches restraint. Governance belongs in the assignments, system design, and demo reviews.

- Classify data before using it: public, internal, sensitive, restricted.
- Require human approval for external sends, public posting, student records, medical/financial data, and administrative action.
- Evaluate every project for hallucination, bias, privacy, prompt injection, and misuse risk.
- Teach audit trails and documentation as core engineering habits.

90-DAY ROADMAP

From funding window to student demos in one semester.

The plan should move fast enough to preserve momentum but structured enough to survive procurement, IT, and faculty load realities.

Weeks 1-2

Confirm budget, room, electrical/HVAC, networking, procurement path, and faculty/staff owner.

Weeks 3-6

Finalize SKUs, order hardware, configure network/storage, install local model stack, prepare pilot curriculum.

RISK CONTROL

The lab succeeds if sustainment is planned on day one.

Hardware purchase is the easy part. Labs fail when ownership, maintenance, curriculum integration, and procurement assumptions are vague.

- Assign at least 0.2 FTE lab ownership in writing.
- Budget annual sustainment: software, support, peripherals, storage, and light refresh.
- Recruit 2-3 student lab assistants from the first cohort.
- Quote final SKUs before promising prices; high-memory hardware pricing is volatile.
- Walk the room with IT before finalizing network, power, cooling, and security assumptions.

NEXT DELIVERABLE

Ask one question, then ship the procurement memo.

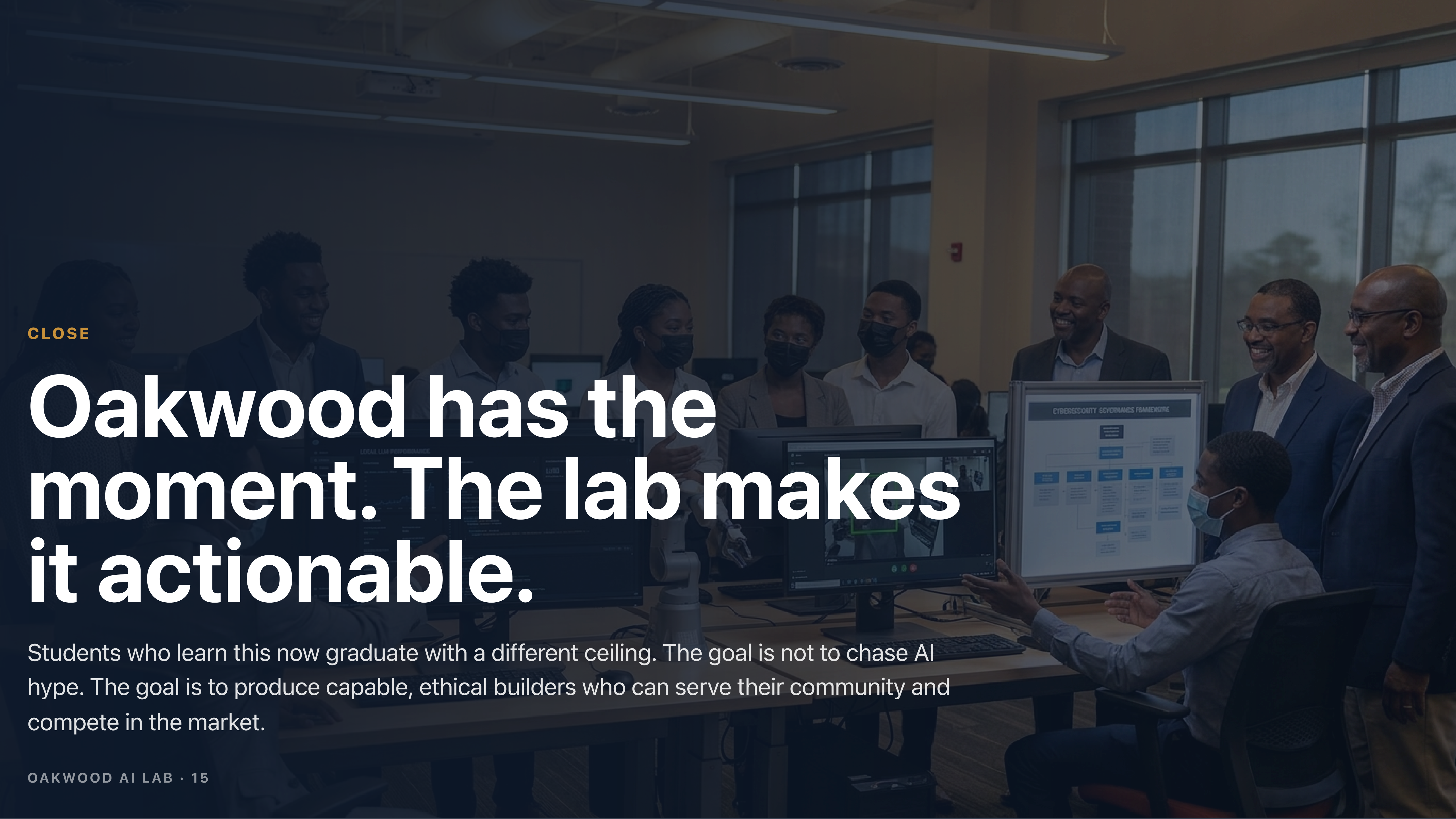
The meeting should produce one concrete answer: which funding window are you working with, and when must the money be committed?

Package A

$20K-$50K: 3-page procurement memo, room layout, quote list, 8-week pilot, lab owner role.

Package B

$100K: Package A plus faculty research framing, employer MOU template, draft grant narrative.



CLOSE

Oakwood has the moment. The lab makes it actionable.

Students who learn this now graduate with a different ceiling. The goal is not to chase AI hype. The goal is to produce capable, ethical builders who can serve their community and compete in the market.

Source and Quality Notes

- Internal context: May 13 Oakwood / Ms. Smith AI Lab note, Jon priority reset map, May 18 working draft, and prior ai-lab-2026 deck.
- Kimi source package contributed the executive structure, investment tiers, roadmap, and strategic styling direction.
- Claims that could not be reliably verified from available public sources were softened or removed, including precise grant portal language and exact volatile pricing claims.
- Official hardware reference used in framing: NVIDIA DGX Spark public page confirms GB10 Grace Blackwell, 128GB coherent unified memory, up to 1 petaFLOP FP4, local models up to 200B parameters, and two-node configurations up to 405B parameters.
- All final hardware budgets are planning estimates and require vendor quotes, room measurements, IT review, and procurement approval.